Cryptography and Network Security: Principles and Practice, 6th Edition, by William Stallings

CHAPTER 10: OTHER PUBLIC-KEY CRYPTOSYSTEMS

TRUE OR FALSE

T F 1. The Diffie-Hellman key exchange is a simple public-key algorithm.

T F 2. The security of ElGamal is based on the difficulty of computing discrete logarithms.

T F 3. For purposes of ECC, elliptic curve arithmetic involves the use of an elliptic curve equation defined over an infinite field.

T F 4. The Diffie-Hellman algorithm depends on the difficulty of computing discrete logarithms for its effectiveness.

T F 5. There is not a computational advantage to using ECC with a shorter key length than a comparably secure TSA.

T F 6. Most of the products and standards that use public-key cryptography for encryption and digital signatures use RSA.

T F 7. ECC is fundamentally easier to explain than either RSA or Diffie-Hellman.

T F 8. A number of public-key ciphers are based on the use of an abelian group.

T F 9. Elliptic curves are ellipses.

T F 10. For determining the security of various elliptic curve ciphers it is of some interest to know the number of points in a finite abelian group defined over an elliptic curve.

T F 11. The form of cubic equation appropriate for cryptographic applications for elliptic curves is somewhat different for GF(2m) than for Zp.

T F 12. An encryption/decryption system requires that point Pm be encrypted as a plaintext.

T F 13. The security of ECC depends on how difficult it is to determine k given kP and P.

T F 14. A considerably larger key size can be used for ECC compared to RSA.

T F 15. Since a symmetric block cipher produces an apparently random output it can serve as the basis of a pseudorandom number generator.

MULTIPLE CHOICE

1. The ___________ protocol enables two users to establish a secret key using a public-key scheme based on discrete logarithms.

   A. Micali-Schnorr B. Elgamal-Fraiser

   C. Diffie-Hellman D. Miller-Rabin

2. _________ can be used to develop a variety of elliptic curve cryptography schemes.

   A. Elliptic curve arithmetic B. Binary curve

   C. Prime curve D. Cubic equation

3. The key exchange protocol is vulnerable to a _________ attack because it does not authenticate the participants.

   A. one-way function B. time complexity

   C. chosen ciphertext D. man-in-the-middle

4. The ________ cryptosystem is used in some form in a number of standards including DSS and S/MIME.

   A. Rabin
   B. Rijnedel
   C. Hillman
   D. ElGamal

5. A(n) ________ is defined by an equation in two variables with coefficients.

   A. abelian group
   B. binary curve
   C. cubic equation
   D. elliptic curve

6. __________ are best for software applications.

   A. Binary curves
   B. Prime curves
   C. Bit operations
   D. Abelian groups

7. An encryption/decryption system requires a point G and an elliptic group _________ as parameters.

   A. Eb(a,q)
   B. Ea(q,b)
   C. En(a,b)
   D. Eq(a,b)

8. For cryptography the variables and coefficients are restricted to elements in a __________ field.

   A. primitive
   B. infinite
   C. public
   D. finite

9. If three points on an elliptic curve lie on a straight line their

sum is __________ .

A. 0 B. 1

C. 6 D. 3

10.____________ makes use of elliptic curves in which the variables and coefficients are all restricted to elements of a finite field.

A. Prime curve B. Elliptic curve cryptography(ECC)

C. abelian group D. Micali-Schnorr

11. For a ___________ defined over GF(2m), the variables and coefficients all take on values in GF(2m) and in calculations are performed over GF(2m).

A. cubic equation B. prime curve

C. binary curve D. abelian group

12. If a secret key is to be used as a _________ for conventional encryption a single number must be generated.

A. discrete logarithm B. prime curve

C. session key D. primitive root

13. The Diffie-Hellman key exchange formula for calculation of a secret key by User A is:

A. K = nB x PA
B. K = nA x PB
C. K = nP x BA
D. K = nA x PA

14. Included in the definition of an elliptic curve is a single element denoted O and called the point at infinity or the __________ .

A. prime point
B. zero point
C. abelian point
D. elliptic point

15. The __________ key exchange involves multiplying pairs of nonzero integers modulo a prime number q. Keys are generated by exponentiation over the group with exponentiation defined as repeated multiplication.

A. Diffie-Hellman
B. Rabin-Miller
C. Micali-Schnorr
D. ElGamal

SHORT ANSWER

1. Elliptic curve arithmetic can be used to develop a variety of elliptic curve cryptography schemes, including key exchange, encryption, and __________ .

2. The purpose of the __________ algorithm is to enable two users to securely exchange a key that can then be used for subsequent encryption of messages.

3. The key exchange protocol vulnerability can be overcome with the use of digital signatures and __________ certificates.

4. The principal attraction of __________, compared to RSA, is that it appears to offer equal security for a far smaller key size, thereby reducing processing overhead.

5. A(n) ___________ G is a set of elements with a binary operation, denoted by *, that associates to each ordered pair (a,b) of elements in G an element (a*b) in G.

6. Two families of elliptic curves are used in cryptographic applications: prime curves over Zp and __________ over GF(2m).

7. We use a cubic equation in which the variables and coefficients all take on values in the set of integers from 0 through p - 1 and in which calculations are performed modulo p for a __________ over Zp.

8. A __________ GF(2m) consists of 2m elements together with addition and multiplication operations that can be defined over polynomials.

9. The addition operation in elliptic curve cryptography is the counterpart of modular multiplication in RSA, and multiple addition is the counterpart of __________ .

10. To form a cryptographic system using __________ we need to find a "hard-problem" corresponding to factoring the product of two primes or taking the discrete logarithm.

11. Eq(a,b) is an elliptic curve with parameters a, b, and q, where _________ is a prime or an integer of the form 2m.

12. The fastest known technique for taking the elliptic curve logarithm is known as the _________ method.

13. Asymmetric algorithms are typically much slower than symmetric algorithms so they are not used to generate open-ended __________ generator bit streams.

14. The __________ pseudorandom number generator is recommended in the ANSI standard X9.82 (Random Number Generation) and in the ISO standard 18031 (Random Bit Generation).

15. The PRNG variable ___________ is defined in NIST SP 800-90 as a number associated with the amount of work required to break a cryptographic algorithm or system.